# Copula Entropy

## Theory and Applications

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# Lecture 4: Theoretical applications (2)

# Copula Entropy: Application 4

Causal Discovery[1]

[1] Jian Ma. "Estimating Transfer Entropy via Copula Entropy". In: *arXiv preprint arXiv:1910.04375* (2019).

# Copula Entropy: Causal Discovery

- Problem
  - To infer causality from time series data by *estimating Transfer Entropy*
- History & Significance
  - Causality is one of the oldest topics in philosophy
  - Causal discovery is a fundamental problem of all sciences
- Correlation vs Causality
  - Correlation does not mean causation
  - Correlation is symmetric while causality is asymmetric
  - Correlation is only helpful for prediction while causality implies temporal relation, intervention and control

# Copula Entropy: Causal Discovery

- **Wiener's Principle**[2]

  Cause should improve the prediction of effect.

[2] Norbert Wiener. "The theory of prediction. Modern mathematics for engineers". In: *New York* 165.6 (1956).

# Copula Entropy: Causal Discovery

- **Granger Causality**[3]

  improvement measured by the variance of prediction error

## Definition (Granger Causality)

Given time series $X_t$, $Y_t$, we say $X$ cause $Y$ in Granger's sense if

$$\delta^2(Y_{t+1}|Y_t, X_t) < \delta^2(Y_{t+1}|Y_t), \quad (1)$$

where $\delta^2$ is variance of prediction error.

[3] Clive WJ Granger. "Investigating causal relations by econometric models and cross-spectral methods". In: *Econometrica: Journal of the Econometric Society* (1969), pp. 424–438, Clive WJ Granger. "Testing for causality: a personal viewpoint". In: *Journal of Economic Dynamics and control* 2 (1980), pp. 329–352.

# Copula Entropy: Causal Discovery

**Transfer Entropy**[4]

- Definition
  improvement on the uncertainty of prediction measured by Shannon entropy

$$TE = \sum p(Y_{t+1}, Y^t, X_t) \log \frac{p(Y_{t+1}|Y^t, X_t)}{p(Y_{t+1}|Y^t)} \tag{2}$$
$$= H(Y_{t+1}|Y^t) - H(Y_{t+1}|Y^t, X_t) \tag{3}$$

[4] Thomas Schreiber. "Measuring information transfer". In: *Physical Review Letters* 85.2 (2000), pp. 461–464.

# Copula Entropy: Causal Discovery

**Transfer Entropy**

- Essentially conditional mutual information

$$TE = I(Y_{t+1}; X_t | Y^t) \tag{4}$$

# Copula Entropy: Causal Discovery

**Transfer Entropy**[5]

- Entropy based estimation

$$TE = H(Y_{t+1}, Y_t) + H(Y_t, X_t) - H(Y_{t+1}, Y_t, X_t) - H(Y_t) \qquad (5)$$

- Issue

  difficult to estimate, some think impossible without model assumptions

5 Terry Bossomaier et al. *An Introduction to Transfer Entropy: Information Flow in Complex Systems*. Springer Cham, 2016.

# Copula Entropy: Causal Discovery

## TE via CE[6]

**Proposition**

$$T_{x\rightarrow y} = -H_c(Y_{t+1}, Y^t, X_t) + H_c(Y_{t+1}, Y^t) + H_c(Y^t, X_t) - H_c(Y^t) \tag{6}$$

**Proof.**

$$\begin{aligned} TE &= \sum p(Y_{t+1}, Y^t, X_t) \log \frac{p(Y_{t+1}|Y^t, X_t)}{p(Y_{t+1}|Y^t)} & (7)\\ &= \sum p(Y_{t+1}, Y^t, X_t) \log \frac{p(Y_{t+1}, Y^t, X_t)p(Y^t)}{p(Y_{t+1}, Y^t)p(Y^t, X_t)} & (8)\\ &= I(Y_{t+1}; Y^t; X_t) - I(Y_{t+1}; Y^t) - I(Y^t; X_t) & (9)\\ &= -H_c(Y_{t+1}; Y^t; X_t) + H_c(Y_{t+1}; Y^t) + H_c(Y^t; X_t) & (10)\\ &= -H_c(Y_{t+1}, Y^t, X_t) + H_c(Y_{t+1}, Y^t) + H_c(Y^t, X_t) - H_c(Y^t) & (11) \end{aligned}$$

□

[6] Jian Ma. "Estimating Transfer Entropy via Copula Entropy". In: *arXiv preprint arXiv:1910.04375* (2019).

# Copula Entropy: Causal Discovery

## TE via CE[7]

**Proposition**

*Transfer Entropy can be represented with only Copula Entropy.*

$$T_{x\rightarrow y} = -H_c(Y_{t+1}, Y^t, X_t) + H_c(Y_{t+1}, Y^t) + H_c(Y^t, X_t) - H_c(Y^t) \quad (12)$$

- Non-parametric Estimator of TE
  1. estimating three or four CE terms in (12);
  2. calculating TE for these estimated CEs.
- inheriting all the merits of the non-parametric CE estimator

[7] Jian Ma. "Estimating Transfer Entropy via Copula Entropy". In: *arXiv preprint arXiv:1910.04375* (2019).

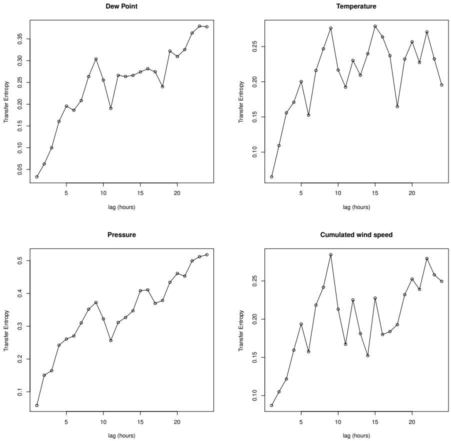
# Copula Entropy: Causal Discovery

**Experiments on the UCI Beijing PM2.5 data**

- Overview of the data
  - Time
    hourly data from 2010-01-01 to 2014-12-31, which results in 43824 samples with missing values.
  - Observations
    - PM2.5 data of US Embassy in Beijing
    - Meteorological data from Beijing Capital International Airport
  - Meteorological factors
    dew point, temperature, pressure, cumulated wind speed, combined wind direction, cumulated hours of snow, cumulated hours of rain.
- Experimental data
  - the first four factors used in the experiments;
  - 1000 samples without missing values (2010-04-02∼2010-05-14).

# Copula Entropy: Causal Discovery

**Results**: Effects of meteorological factors on PM2.5




- Two phrases
  - Sharp increase phrase: the first 9 hours time lag, and peak at about 9 hours lag;
  - Flat increase phrase: TE of Dew point and pressure increase with relatively flat rate while TE of temp. and cumulated wind speed does increase any more.
- Interpretation
  - The effects do not show immediately and are cumulating processes.

# Copula Entropy: Application 5

Time Lag Estimation[8]

[8] Jian Ma. "Identifying Time Lag in Dynamical Systems with Copula Entropy based Transfer Entropy". In: *arXiv preprint arXiv:2301.06037*, arXiv:2301.06037 (2023). arXiv: 2301.06037 [cs.LG].

# Copula Entropy: Time Lag Estimation

- Problem
  - To identify time lag in dynamical systems with copula entropy based transfer entropy
- Significance
  - Time lag is ubiquitous in physical, social, and biological systems
  - Identifying time lag is of fundamental importance in applications of dynamical systems
- Related Methods
  - Auto-correlation
  - Time-delayed mutual information

# Copula Entropy: Time Lag Estimation

- System model

$$y_t = f(x_{t-l}, \mathbf{z}_t) + \epsilon \tag{13}$$

- Our method
  1. estimating transfer entropies in time lag window from data with the CE-based estimator
  2. identifying the time lag $l$ associated with the maximal TE value

$$\hat{l} = \arg\max_{l} TE_{X\rightarrow Y}(l) \tag{14}$$

# Copula Entropy: Time Lag Estimation

**Simulations**

1. generate trajectories from five simulated dynamical system with respect to different state or ouput lags
2. identify the time lag with our method

# Copula Entropy: Time Lag Estimation

**Simulated systems**

1. a system driven by random input with output lag

$$\begin{aligned} x_i &= \xi_1, \\ y_{i+l} &= x_i + \xi_2; \end{aligned} \tag{15}$$

2. a system driven by sine function with output lag

$$\begin{aligned} x_i &= \sin(2\pi i/m) + \xi_1, \\ y_{i+l} &= x_i + \xi_2; \end{aligned} \tag{16}$$

3. Wiener process with output lag

$$\begin{aligned} x_i &= x_{i-1} + \xi_1, \\ y_{i+l} &= x_i + \xi_2; \end{aligned} \tag{17}$$

# Copula Entropy: Time Lag Estimation

**Simulated systems**

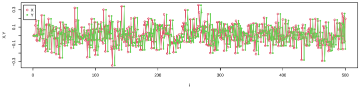
4. a first-order linear system with state lag

$$x_i = \alpha x_{i-1} + \beta x_{i-l} + \xi_1, \\ y_i = x_i + \xi_2; \tag{18}$$

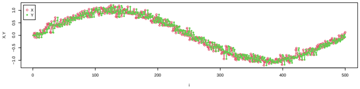
5. a first-order nonlinear system with state lag

$$x_i = \alpha x_{i-1} + \beta x_{i-l} + \xi_1, \\ y_i = x_i^2 + \sin(x_i) + \xi_2. \tag{19}$$

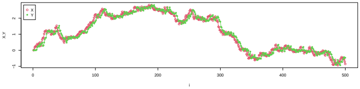
# Copula Entropy: Time Lag Estimation



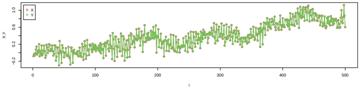

(a) random system



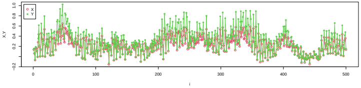

(b) sin function




(c) Wiener process



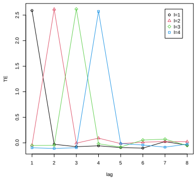

(d) linear first-order system



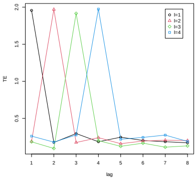

(e) nonlinear first-order system

Figure: Simulated trajectories.

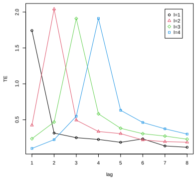
# Copula Entropy: Time Lag Estimation



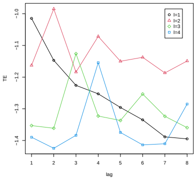

(a) random system



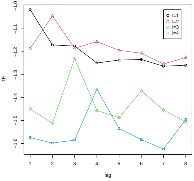

(b) sin function




(c) Wiener process




(d) linear system



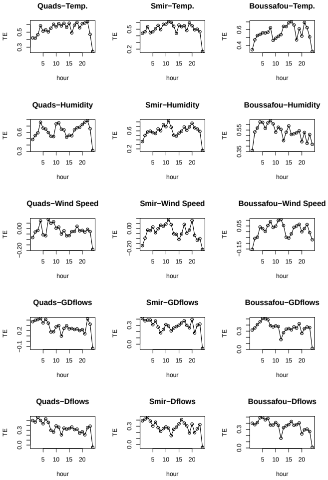

(e) nonlinear system

Figure: Simulation results of the experiments on time lag estimation.

# Copula Entropy: Time Lag Estimation

**Real data: Power consumption of the Tetouan city**[9]

- Data
  - power consumption of 3 networks in 2017
  - weather factors, including temperature, humidity, wind speed, general diffuse flows, diffuse flows
- Power consumption forecast
  - To identify time lags from weather to power consumption




[9] **Asuncion2007**.

# Copula Entropy: Application 6

System Identification[10]

[10] Jian Ma. "Two-Sample Test with Copula Entropy". In: *arXiv preprint arXiv:2307.07247* (2023). arXiv: 2307.07247 [stat.ME].

# Copula Entropy: System Identification

- Problem
  - To discover differential equation from time series data
- Significance
  - differential equations are the main mathematical tools for modeling dynamical systems
  - discovering differential equations of dynamical systems has wide applications in many scientific fields
- Related Methods
  - SINDy
  - Gaussian processes

# Copula Entropy: System Identification

## Sparse Identification of Nonlinear Dynamical systems (SINDy)[11]

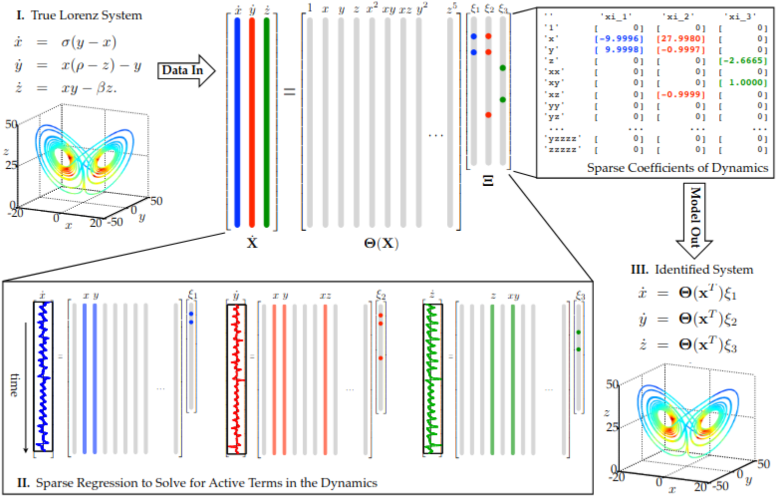
- Idea

  use **sparse regression** to find a sparse linear combination of basis functions that best approximate the differential equations of dynamical system
  $$\frac{dx}{dt} = f(x(t)). \tag{20}$$
- Method
  1. perform numerical differentiation $\frac{dx}{dt}$;
  2. specify a set of basic functions $\Theta(x)$ such that
  $$\frac{dx}{dt} = \Theta(x)\xi; \tag{21}$$
  3. LASSO regression
  $$\hat{\xi} = \arg\min_{\xi} ||\Theta(x)\xi - \frac{dx}{dt}|| + \lambda||\xi||_1 \tag{22}$$

[11] Steven L Brunton, Joshua L Proctor, and J Nathan Kutz. "Discovering governing equations from data by sparse identification of nonlinear dynamical systems". In: *Proceedings of the national academy of sciences* 113.15 (2016), pp. 3932–3937.

# Copula Entropy: System Identification

## Sparse Identification of Nonlinear Dynamical systems (SINDy)



# Copula Entropy: System Identification

- Our method
  1. calculating the derivative of system variables with differential operator;
  $$\left.\frac{dx}{dt}\right|_{t=t_0} \approx \frac{x_{t_1} - x_{t_0}}{t_1 - t_0}. \tag{23}$$
  2. estimating the CEs $H_c(\cdot, \cdot)$ between the calculated derivatives $\frac{dx}{dt}$ and the covariates of the system $x, y, z, \ldots$;
  3. selecting the covariates with high CE value for each derivatives $\frac{dx}{dt}$.

# Copula Entropy: System Identification

**Simulations**

simulating time series data from Lorenz system and Rössler system

**Lorenz system**

$$\frac{dx}{dt} = \sigma(y - x) \quad (24)$$

$$\frac{dy}{dt} = \rho x - y - xz \quad (25)$$

$$\frac{dz}{dt} = -\beta z + xy \quad (26)$$

**Rössler system**

$$\frac{dx}{dt} = -(y + z) \quad (27)$$

$$\frac{dy}{dt} = x + ay \quad (28)$$

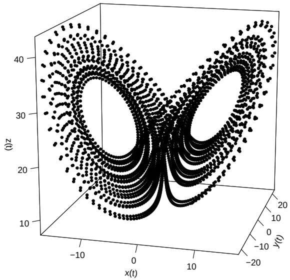
$$\frac{dz}{dt} = b + z(x - c) \quad (29)$$

# Copula Entropy: System Identification

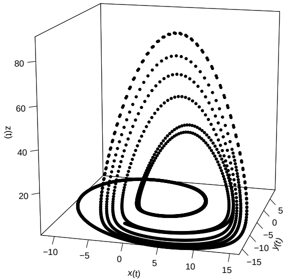
**Simulations**

simulating time series data from Lorenz system and Rössler system

**Lorenz system**




**Rössler system**



# Copula Entropy: System Identification

## Results - Lorenz system

$$\frac{dx}{dt} = 10(y - x)$$
$$\frac{dy}{dt} = 28x - y - xz$$
$$\frac{dz}{dt} = -\frac{8}{3}z + xy$$

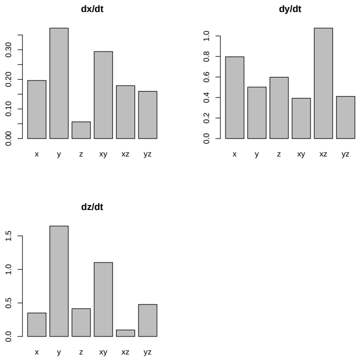



Figure: Results.

# Copula Entropy: System Identification

**Results - Rössler system**

$$\frac{dx}{dt} = -(y+z)$$
$$\frac{dy}{dt} = x + 0.38y$$
$$\frac{dz}{dt} = 0.2 + z(x-5.7)$$

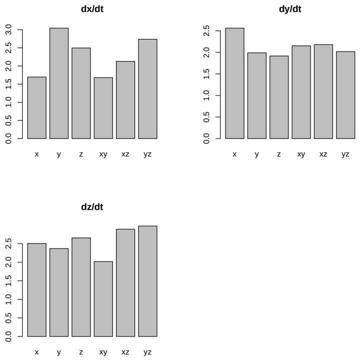



Figure: Results.

# Monograph

Jian Ma. “Copula Entropy: Theory and Applications”. In: *arXiv preprint arXiv:2025.18168* (2025)

# Thanks!

Q&A